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Graduation Project II
Smart industrial distribution board
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Abstract

Traditional industrial power systems often face severe vulnerabilities due to separate, uncoordinated protection mechanisms, lack of diagnostic tracking, and volatile telemetry displays that complicate manual data reading. These inefficiencies lead to prolonged electrical hazards, catastrophic equipment failures, and extended maintenance downtime. To solve these critical challenges, a Smart Industrial Distribution Board was developed as an integrated, intelligent, and highly reliable alternative to conventional panel boards.

The proposed system seamlessly incorporated five core functionalities: a smart 3-in-1 protection mechanism against overcurrent, overvoltage, and undervoltage; a "Black Box" diagnostics module for rapid post-fault analysis; software-filtered stable display readings; independent multi-load management; and a highly cost-effective hardware architecture. This solution was physically implemented using an Arduino Uno controller as the central processing unit, integrated with precise voltage and current sensors to actively track electrical parameters. The physical layout, featuring the external operator controls, pushbuttons, and the integrated LCD control screen, was structured as detailed. Internally, the microcontroller was wired to a 4-channel relay array, circuit breakers, a voltage regulator, and an active cooling fan to maintain thermal stability, as documented in the internal wiring diagram.

The experimental results demonstrated exceptional operational efficiency. The integrated system successfully achieved a rapid fault-detection and isolation response time of less than 150 milliseconds during simulated overcurrent and voltage anomalies, while the display smoothing algorithm effectively eliminated screen flicker to ensure high readability. In conclusion, the project successfully validated that incorporating intelligent microcontroller-based diagnostics and protection into industrial distribution infrastructure significantly minimizes critical downtime, streamlines fault isolation, and delivers a robust, scalable, and economically viable alternative for modern smart-grid environments.

Keywords:

1. Smart Industrial Distribution Board
2. 3-in-1 Intelligent Protection
3. Black Box Diagnostics
4. Arduino Uno Controller
5. Multi-Load Management
6. Fault Isolation Response Time
7. Cost-Effective Automation



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List of Abbreviations

Abbreviation	Definition
SUT	El Sewedy University of Technology
ABS	Acrylonitrile Butadiene Styrene (a common thermoplastic polymer).
AC	Alternating Current.
ADC	Analog-to-Digital Converter.
BOM	Bill of Materials.
DC	Direct Current.
DIN	Deutsches Institut für Normung (German Institute for Standardization, commonly referring to mounting rail standards).
DSP	Digital Signal Processing.
DSO	Digital Storage Oscilloscope.
EGP	Egyptian Pound (currency).
EMI	Electromagnetic Interference.
ETU	Electronic Trip Unit.
FFT	Fast Fourier Transform.
FIFO	First-In, First-Out (a queue/buffer data structure).
I2C	Inter-Integrated Circuit (a synchronous serial communication protocol).
IEC	International Electrotechnical Commission.
IIoT	Industrial Internet of Things.
ISO	International Organization for Standardization.
LCD	Liquid Crystal Display.



Abbreviation	Definition
MQTT	Message Queuing Telemetry Transport.
NILM	Non-Intrusive Load Monitoring.
OS	Operating System.
PC	Program Counter (microcontroller register).
PCB	Printed Circuit Board.
PLC	Programmable Logic Controller.
POST	Power-On Self-Test.
RC	Resistor-Capacitor (circuit).
RMS	Root Mean Square.
SCADA	Supervisory Control and Data Acquisition.
SME	Small-to-Medium Enterprise.
SNR	Signal-to-Noise Ratio.
SRAM	Static Random Access Memory.
SWOT	Strengths, Weaknesses, Opportunities, and Threats.
VCC	Voltage Common Collector (positive supply voltage).

Chapter 1: Introduction

1.1 Background and Context

The modern industrial sector relies entirely on continuous, high-quality electrical power to drive manufacturing lines, heavy machinery, and automated systems. Within this domain, electrical distribution boards serve as the critical control nodes responsible for routing electrical energy from main supply lines down to individual localized loads. In standard practice, these panels depend primarily on passive, electromechanical switchgear—such as traditional thermal-magnetic circuit breakers and physical fuses—to protect infrastructure. While these conventional devices excel at isolating extreme, sustained overcurrent events, they operate as isolated, analog components. They do not possess the capability to process real-time data, communicate telemetry, or adapt dynamically to fluctuating grid conditions.

To bridge this technological gap, the field of electrical engineering is rapidly pivoting toward smart grids and digital power management. A non-specialist must understand a few key terms to follow this evolution: *overcurrent* refers to a hazardous situation where electrical current exceeds the rated capacity of a circuit; *overvoltage* and *undervoltage* describe supply voltage levels that rise too high or drop too low, both of which degrade electronic components; and *telemetry* represents the automated measurement and wireless transmission of digital data from remote sources.

While heavy industries often deploy expensive Programmable Logic Controllers (PLCs) to achieve smart automation, small-to-medium manufacturing operations remain stuck with completely unmonitored legacy systems. This stark division leaves standard panel boards entirely blind to transient power quality issues and post-fault diagnostics. Consequently, when a failure occurs, facilities suffer from extended operational downtime simply trying to identify the root cause. This project directly addresses this systemic vulnerability by developing an intelligent, digitally integrated, and highly accessible distribution node designed specifically to eliminate these diagnostic and protective blind spots.



1.2 Problem Statement

Traditional industrial power distribution networks suffer from widespread inefficiencies, safety vulnerabilities, and high maintenance overheads due to several core architectural limitations:

- **Fragmented and Uncoordinated Protection:** Conventional panels separate protection mechanisms across independent hardware units (such as standalone thermal-magnetic circuit breakers, overvoltage relays, and phase-sequence monitors). This lack of coordination increases panel space requirements, complicates control wiring, and results in slower, uncoordinated trip responses.
- **The "Blind Spot" in Post-Fault Diagnostics:** When a conventional protective device trips, plant technicians are left with no contextual data regarding the precise cause of the failure. Determining whether a shutdown was triggered by a transient overvoltage spike, a sustained thermal overcurrent, or a localized short circuit requires manual troubleshooting with multimeters, leading to extended maintenance downtime.
- **Signal Noise and Volatile Telemetry Displays:** Industrial environments are plagued by heavy electromagnetic interference (EMI) and transient harmonics caused by inductive loads, variable frequency drives, and switching power supplies. When raw sensor data is directly fed to control screens, it results in severe reading fluctuations and high-frequency screen flicker, which degrades display readability and compromises manual data tracking.
- **Rigid, Unified Load Control:** Standard distribution boards lack granular control over individual branches. If a minor fault occurs on a non-critical load, the entire panel or a large subsection may trip upstream, resulting in unnecessary operational disruptions for neighboring, healthy industrial processes.
- **Prohibitive Cost of Smart Automation:** Commercial intelligent distribution solutions, such as high-end Programmable Logic Controllers (PLCs) and networked power meters, carry prohibitive capital costs and require complex programming architectures, rendering them economically unviable for small-to-medium enterprises (SMEs) or localized industrial workshops.



1.3 Motivation and Justification

The urgency to modernize industrial power infrastructure has reached a critical tipping point. As global manufacturing rapidly transitions toward highly automated, interconnected systems—often dubbed Industry 4.0—the tolerance for electrical instability has dropped to near zero. Modern production lines rely heavily on sensitive microprocessor-based machinery, robotics, and digital controllers. While these advancements drastically improve efficiency, they are vastly more vulnerable to minor power anomalies than the rugged, purely mechanical equipment of the past.

Operating under legacy power distribution methods is no longer financially or operationally viable. According to recent industrial survey data, unexpected downtime costs manufacturers an estimated \$50 billion annually, with power quality anomalies and uncoordinated protection systems ranking as leading root causes. For small-to-medium enterprises (SMEs), a single prolonged power outage or an undiagnosed electrical fault can result in catastrophic revenue losses, supply chain bottlenecks, and severe equipment degradation.

Resolving this issue now is also driven by the global push for energy efficiency and resource optimization. Traditional, "blind" distribution boxes offer no data insight, preventing facilities from identifying energy waste, phase imbalances, or failing components before they trigger a catastrophic system shutdown. Resolving these diagnostic and protective vulnerabilities through localized, cost-effective digital intelligence is not just a minor hardware upgrade; it is an immediate economic necessity required to protect industrial assets, ensure worker safety, and prevent widespread operational disruptions in an increasingly competitive global market.



1.4 Project Objectives

To successfully address the identified industrial vulnerabilities and validate the performance of the proposed system, the project is structured around the following specific, measurable engineering objectives:

- **Design** a unified, software-defined 3-in-1 protection algorithm capable of simultaneously monitoring overcurrent, overvoltage, and undervoltage anomalies, ensuring the system can trigger a safe isolation sequence within a target response window of less than 150 milliseconds.
- **Develop** an autonomous embedded "Black Box" diagnostic routine that captures and freezes critical telemetry parameters (voltage, current, and active load states) exactly at the moment of a trip event, eliminating diagnostic blind spots for maintenance technicians.
- **Implement** a digital signal-smoothing and filtering algorithm within the microcontroller firmware to damp high-frequency electrical noise, achieving stable, fluctuation-free real-time parameter readings on the localized control screen interface.
- **Evaluate** the operational independence of a multi-load management matrix by executing individual branch isolation under simulated fault conditions, verifying that localized failures do not cause cascading shutdowns on adjacent, healthy industrial channels.
- **Compare** the overall material cost and manufacturing complexity of the engineered prototype against standard commercial alternatives (such as low-end PLCs or commercial digital protection relays) to prove its financial viability as a cost-effective solution for small-to-medium enterprises.



1.5 Scope and Limitations

To ensure a structured engineering lifecycle and protect the development process from scope creep, the operational boundaries, technical constraints, and deliberate exclusions of this project are strictly defined as follows:

What IS Included:

- **Hardware Architecture:** The prototype is entirely bounded by an 8-bit microcontroller framework utilizing the Arduino Uno platform. It includes the physical integration of analog current and voltage sensors, a 4-channel relay array for load control, industrial circuit breakers for hardware backup protection, an active cooling system, and an LCD control screen with manual pushbutton overrides.
- **Functional Capabilities:** The system is constrained to executing five core embedded routines: digital 3-in-1 protection loops (overcurrent, overvoltage, undervoltage), automated "Black Box" memory capture during faults, software-based telemetry signal filtering, and independent routing for up to three localized sub-loads.
- **Testing and Time Period:** Evaluation is restricted to localized laboratory simulations and controlled benchmark testing conducted within the 2025–2026 academic project cycle.

What IS NOT Included:

- **Industrial Protocols and High-Power Hardware:** This project deliberately excludes high-voltage or three-phase heavy industrial power integration; it operates strictly as a low-voltage, single-phase prototype. It does not include standard industrial communication protocols such as Modbus, Profibus, or CAN bus.
- **IoT and Remote Connectivity:** Networked wireless communications, cloud-based logging, mobile applications, and remote internet-of-things (IoT) dashboards are entirely out of scope for this iteration.
- **Advanced Automation Enclosures:** The project does not feature full-scale Programmable Logic Controller (PLC) networking, Supervisory Control and Data Acquisition (SCADA) integration, or formal NEMA/IP-rated industrial climate enclosures. All operations are confined to localized, standalone automated control.



1.6 Project Outcome and Deliverables

The successful execution of this engineering project culminates in a set of distinct, tangible deliverables that span hardware construction, embedded software development, and technical documentation. The final outputs of this project include:

- **Fully Functional Hardware Prototype:** A complete, low-voltage smart distribution board enclosure integrated as a standalone operational unit. This physical deliverable incorporates an Arduino Uno controller, an LCD control screen, external status indicators, manual load pushbuttons, a 4-channel relay array, a voltage/current sensor network, primary circuit breakers, a voltage regulator, and an active cooling system.
- **Embedded Control Firmware:** The complete, fully commented production-ready source code written in C/C++ within the Arduino IDE. This software asset houses the central processing loop, including the 3-in-1 protective thresholds, the mathematical signal-filtering algorithms for telemetry display stabilization, and the volatile memory-freezing routine for the "Black Box" diagnostics module.
- **Comprehensive Project Book and Documentation:** A highly detailed, multi-page technical report containing comprehensive electrical schematics, internal wiring topographies, software flowcharts, component bills of materials (BOM), and rigorous experimental performance validation matrices.
- **Empirical Calibration and Testing Dataset:** A structured collection of laboratory test data log sheets that record exact sensor calibration curves, simulated fault scenarios, and microsecond-accurate hardware response times across a variety of overcurrent and voltage variance tests.
- **Demonstration Video:** A high-definition recorded multimedia presentation demonstrating the physical system under normal and simulated fault conditions. The video explicitly showcases real-time display smoothing, independent three-load routing, instantaneous automated fault isolation, and the post-fault retrieval of diagnostic parameters from the system's "Black Box" state.



Chapter 2: Literature Review

2.1 Theoretical Background

This chapter establishes the core theoretical principles, mathematical frameworks, and signal processing algorithms that govern the operation of the Smart Industrial Distribution Board. Understanding these underlying concepts is essential to analyzing how the system transitions from raw analog sensor data to deterministic protective and diagnostic actions.

3.1 AC Power Fundamentals and Sensor Theory

To accurately monitor and protect an electrical circuit, the embedded system must sample and process time-varying alternating current (AC) waveforms.

Alternating Current Waveforms

A pure AC voltage or current waveform is modeled mathematically as a sinusoidal function of time:

$$v(t) = V_m \sin(\omega t + \theta)$$

Where V_m is the peak amplitude, $\omega = 2\pi f$ is the angular frequency (with a standard grid frequency $f = 50 \text{ Hz}$ or 60 Hz) and θ is the initial phase angle. Because the instantaneous value of a sine wave continuously alternates and averages out to zero over a complete cycle, electrical systems rely on the **Root Mean Square (RMS)** value to express the effective voltage or current. The RMS value represents the equivalent DC value that would deliver the same average thermal power to a purely resistive load. For a pure sinusoidal wave, the calculation is defined as:

$$V_{RMS} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt} = \frac{V_m}{\sqrt{2}} \approx 0.707V_m$$

The microcontroller must perform a discrete approximation of this continuous integral by rapidly taking a finite number of samples (N) over exactly one or more complete waveforms:

$$V_{RMS} \approx \sqrt{\frac{1}{N} \sum_{i=1}^N v_i^2}$$



Hall-Effect Current Sensing

For non-invasive current tracking, the system builds on the principles of **Lorentz Force** and the **Hall Effect**. When an electrical current passes through a conducting material placed within a magnetic field, the magnetic field exerts a transverse force on the moving charge carriers. This force drives the charges toward one side of the conductor, creating a measurable potential difference across the material known as the Hall Voltage (V_H)

In practical system implementation, a linear Hall-effect current sensor (such as the ACS712) routes the load current directly through an internal copper conduction path near an embedded Hall-effect IC. The flowing current generates a localized magnetic field proportional to the current magnitude. The internal IC converts this magnetic field into a proportional analog voltage output. The microcontroller samples this analog voltage, maps it against the sensor's baseline sensitivity rating (typically expressed in millivolts per Ampere, mV/A, and extracts the instantaneous AC current value.

Electrical Fault Topologies and Protection Theory

The system replaces independent mechanical trip mechanisms with a unified, software-defined protection loop designed to mitigate three critical electrical fault conditions.

Overcurrent Principles

Overcurrent occurs when the current passing through a circuit exceeds its rated thermal capacity. It generally falls into two distinct engineering sub-categories:

- **Overload:** A minor, sustained current excess caused by connecting too many loads or mechanical binding in motorized equipment. Overloads generate cumulative heat over time, threatening insulation integrity.
- **Short Circuit:** A catastrophic, near-instantaneous current spike caused by an accidental zero-resistance connection between live lines or line-to-earth.

Traditional circuit breakers rely on a bimetallic strip that warps under slow thermal accumulation, alongside an electromagnetic coil that trips instantly during short circuits. The smart distribution board replicates this dual-stage response entirely via software logic, protecting wires from thermal failure while preventing nuisance trips during brief startup current inrushes.

Overvoltage Principles

Overvoltage describes a scenario where the supply voltage climbs significantly above the nominal grid rating (e.g., exceeding 242 V on a standard 220 V single-phase line).



Overvoltage anomalies are caused by utility switching operations, lightning strikes, or sudden major load rejections within the industrial plant. This condition forces excessive voltage stress across internal insulations and electronic components, causing dielectric breakdown, insulation puncturing, and catastrophic hardware destruction.

Conversely, undervoltage (or a "brownout") occurs when the line voltage drops below acceptable functional thresholds (e.g., dropping below 198 V on a 220 V system).

Undervoltage states typically arise from heavily overloaded local grids, phase imbalances, or the startup current draw of massive industrial motors upstream. Undervoltage is highly destructive to inductive industrial machinery; as the voltage drops, electric motors must pull significantly higher currents to maintain their mechanical torque output $P = V \times I \times \cos \phi$. This secondary current surge triggers rapid overheating within the motor windings.

Embedded Digital Signal Filtering Algorithms

Industrial facilities are notoriously harsh environments for sensitive microelectronics due to severe **Electromagnetic Interference (EMI)**. The rapid switching of high-power contactors, variable frequency drives, and inductive loads injects high-frequency noise spikes and transient harmonics directly into sensor signal lines. If these raw, noisy signals are routed unfiltered to a control screen, it results in rapid, unreadable display fluctuations and can trigger dangerous false protective trips.

To eliminate this without adding expensive analog hardware filters, the system builds on discrete time-domain digital signal processing (DSP). It implements a software-defined **Moving Average Filter** (also known as a rolling mean filter). This algorithm maintains a running buffer of the last M consecutive analog sensory readings. Every time a new sample is collected, the oldest sample is discarded from the array, and a new average is calculated. The mathematical expression for the moving average filter in the discrete time domain is:

$$y[n] = \frac{1}{M} \sum_{i=0}^{M-1} x[n-i]$$

Where $x[n]$ represents the latest raw analog sample input, $y[n]$ represents the smoothed output value sent to the display logic, and M represents the window size (or filter depth). By increasing M , the system acts as a digital low-pass filter, attenuating high-frequency noise and stabilizing telemetry readouts. The filter window size must be carefully balanced via code optimization: a large M yields exceptional screen stability but introduces a time delay into the control loop, whereas a small M preserves near-instantaneous response times at the expense of higher noise vulnerability.



The "Black Box" State-Freezing Principle

The diagnostic core of this project relies on a state-freezing software routine modeled after commercial aviation flight data recorders. In an ordinary panel board, once a circuit breaker trips, the dynamic parameters leading up to the failure are completely lost, leaving technicians with zero context.

The "Black Box" principle implemented in this system breaks the control loop down into a high-speed deterministic conditional loop. During normal operations, the microcontroller continuously overwrites local registers with volatile operational metrics: active voltage, load current draws, active time-stamps, and load states.

The moment any software-defined protective threshold (Overcurrent, Overvoltage, Undervoltage) is breached, the embedded logic executes an un-interruptible sequence:

1. It immediately drives the respective relay pins low to isolate the affected electrical channel.
2. It abruptly freezes the data-logging register updates, permanently preserving the final "pre-trip state" vector ($[V, I, t \text{ Fault Type}]$)
3. It routes this frozen snapshot to the localized visual display interface and sound alarms.

2.2 Related Work

Hardware-Centric Switchgear Methods

These solutions rely heavily on specialized, localized electronic modules and industrial-grade automation controllers to execute rigid, hardware-level protection and monitoring.

- **Approach 1: Molded Case Circuit Breakers (MCCBs) with Integrated Electronic Trip Units (ETUs)**
 - *What they did:* Industrial manufacturers integrated dedicated microprocessing trip units directly into solid-state circuit breakers to replace legacy bimetallic strips and magnetic solenoids. These units continuously sample incoming current waveforms using internal current transformers.
 - *What they achieved:* This approach achieved highly precise, programmable Long-Time, Short-Time, and Instantaneous (LSI) trip curves, significantly reducing nuisance trips and minimizing overcurrent damage to heavy factory machinery.
 - *Limitations:* These systems remain completely closed, proprietary, and isolated. They do not track voltage anomalies (such as undervoltage) within the same unit, lack multi-load



branch separation, and offer no local post-fault diagnostic registers to freeze parameters for plant operators.

- **Approach 2: Programmable Logic Controller (PLC) Managed Panels**

- *What they did:* Researchers and automation engineers deployed industrial PLCs (such as the Siemens S7-1200) paired with external analog voltage and current transducers to supervise low-voltage distribution lines.
- *What they achieved:* They successfully built a highly rugged, deterministic protection panel with complete noise immunity to electromagnetic interference (EMI) and a localized Human-Machine Interface (HMI) for real-time data viewing.
- *Limitations:* PLC-based implementations suffer from prohibitive capital costs and expensive software licensing fees. Furthermore, they feature highly complex control wiring and require specialized programming expertise, making them completely unviable for small-to-medium enterprises (SMEs) or localized workshops.

IoT-Connected Smart Energy Networks

This class of solutions leverages network-enabled microcontrollers and cloud computing infrastructures to provide wide-area remote telemetry tracking and logging.

- **Approach 1: Cloud-Connected ESP32 Energy Metering Nodes**

- *What they did:* Researchers engineered smart energy monitors using Wi-Fi-enabled microcontrollers (such as the ESP32) integrated with dedicated metering integrated circuits (ICs) like the ADE7753 to stream power parameters to external servers.
- *What they achieved:* They achieved unparalleled remote visibility, allowing plant managers to track real-time power consumption, voltage drops, and accumulated energy logs from centralized web dashboards via MQTT protocols.
- *Limitations:* These configurations introduce severe vulnerabilities regarding network latency and wireless signal degradation inside dense, metallic factory environments. More importantly, cloud-dependent systems fail to guarantee the localized, sub-millisecond deterministic processing speeds required to safely isolate destructive overcurrent or overvoltage faults when a network connection drops.

- **Approach 2: Raspberry Pi Gateways with Local Database Logging**

- *What they did:* Single-board computers were integrated into distribution boxes to sample sensor data via SPI/I2C communication channels and log metrics directly into local databases (such as InfluxDB).



- *What they achieved:* This methodology provided high-capacity local historical logging, exceptional data visualization, and streamlined structural data reporting without relying on a continuous internet connection.
- *Limitations:* Microprocessors like the Raspberry Pi run non-real-time operating systems (such as Linux), which are highly prone to operating system freezes, ungraceful corruption during sudden power failures, and slow boot times. They lack the high-speed deterministic input/output (I/O) control required to manage rapid relay isolation vectors.

Advanced AI-Based Predictive Platforms

These forward-looking research concepts integrate machine learning and edge-computing models to predict faults before they manifest physically.

- **Approach 1: Non-Intrusive Load Monitoring (NILM) with Machine Learning**

- *What they did:* Researchers deployed artificial neural networks (ANNs) and support vector machines (SVMs) at a single centralized monitoring node to disaggregate the total current waveform of a facility into individual load signatures.
- *What they achieved:* They achieved high accuracy in tracking individual machine operating states and identifying specific appliance efficiencies from a singular point of installation without placing sensors on every branch.
- *Limitations:* NILM algorithms demand immense computational processing power, which requires high-end edge AI hardware or external server processing. Additionally, these systems struggle heavily with high-frequency industrial noise, require massive training datasets, and cannot execute instantaneous safety-critical tripping maneuvers.

- **Approach 2: Embedded Tiny ML Predictive Maintenance Modules**

- *What they did:* Engineers compiled highly optimized, compressed machine learning models (Tiny ML) directly onto 32-bit microcontrollers to analyze current waveform distortions and predict winding insulation breakdown in downstream induction motors.
- *What they achieved:* This approach successfully transformed distribution boards from reactive protective devices into proactive maintenance advisors, isolating failing loads before a hard short-circuit event occurred.
- *Limitations:* Tiny ML implementations are restricted by strict hardware constraints, requiring meticulous memory optimization to avoid stack overflows. They are highly complex to calibrate and lack a generalized, scalable architecture, meaning a model trained for a specific motor cannot easily protect general multi-load networks.



2.3 Comparative Analysis

Reference / Method	Application	Key Result	Limitation
Electronic Trip Units (ETU)	High-capacity industrial feeders	High-precision protection curves.	Cost-prohibitive; lacks voltage-based fault diagnostics.
PLC-SCADA Panels	Heavy manufacturing	High EMI immunity; deterministic control.	Extremely large footprint; requires proprietary programming.
Cloud-Connected IoT Nodes	Residential smart metering	Excellent remote data streaming via MQTT.	High latency; dependent on network connectivity for safety.
Raspberry Pi Gateways	Commercial energy logging	High-capacity data visualization.	Non-real-time OS; prone to crashes during power drops.
NILM Algorithms	Energy disaggregation	Identifies appliances from aggregate load.	Demands excessive CPU power; cannot execute fast trips.
TinyML Modules	Predictive motor maintenance	Detects winding breakdown early.	Restricted memory; lacks generalized network architecture.
Proposed Smart Board	Small-scale industrial distribution	3-in-1 protection + Offline "Black Box" logging.	Limited to single-phase low-voltage grids (currently).

2.4 Research Gap and Contribution

A critical synthesis of the reviewed literature reveals a pronounced technological gap at the intersection of power systems engineering and low-cost embedded automation. While commercial switchgear manufacturers have successfully brought ultra-high-end, cost-prohibitive PLC-SCADA panels to market for heavy industries, and academic researchers have extensively published vulnerable, cloud-dependent IoT energy meters for light commercial or residential tracking, there is a distinct absence of a localized, autonomous, and economically accessible smart protection node designed specifically for small-to-medium industrial operations.



Specifically, the existing literature and current market state of practice fail to address the following critical areas that this project directly resolves:

- **Unfiltered Telemetry and Display Fluctuation:** Existing low-cost microcontroller prototypes published in recent literature widely overlook the practical constraints of industrial control environments. They routinely feed raw sensor data directly to visual interfaces, failing to account for high-frequency switching noise and electromagnetic interference (EMI) that render screens unreadable. This project fills that gap by implementing an optimized software-defined moving average filter that damps signal fluctuations to guarantee telemetry stability.
- **Network-Dependent or Lost Diagnostic Trajectories:** Contemporary smart metering research heavily relies on continuous cloud connectivity or external memory shields to log historical parameters. If an internet connection drops or a sudden power failure occurs, the data is permanently lost. The existing literature does not provide an offline, zero-overhead data-freezing mechanism. This project fills this exact gap by introducing an embedded "Black Box" routine that operates entirely on localized microcontroller registers, freezing and isolating the pre-trip state vector at the exact microsecond of a fault without network dependencies.
- **Fragmented Protective Cohesion vs. Multi-Load Independence:** Low-cost prototypes in literature generally focus on a singular parameter (e.g., current monitoring only) or treat the entire panel as a single branch, leading to uncoordinated tripping and widespread cascading power shutdowns across healthy production channels. This project fills this gap by integrating a cohesive, 3-in-1 software-defined protection loop (overcurrent, overvoltage, undervoltage) and mapping it across a multi-channel relay matrix to execute deterministic, independent multi-load isolation.

By designing, prototyping, and validating an integrated system that merges these three computational layers onto a single, open-source microprocessing architecture, this project moves past the fragmentation of traditional switchgear and the vulnerabilities of cloud-reliant instrumentation. It provides a robust, self-contained, and financially viable smart distribution node that fills a vital operational vacancy in modern localized grid networks.



Chapter 3: Feasibility Study

3.1 Technical Feasibility

To confirm that the Smart Industrial Distribution Board is a realistic engineering project rather than just a theoretical concept, its implementation was evaluated based on three critical parameters: hardware/software availability, team technical skills, and compatibility with existing industrial environments.

1. Hardware and Software Resource Availability

Every component required to build, program, and house the prototype is commercially mature, mass-produced, and highly accessible.

- **Hardware Availability:** The core processing unit—the Arduino Uno—is widely available globally along with standard analog voltage transformers and non-invasive current sensors (such as the ACS712 or split-core current transformers). The actuation hardware relies on standard isolated 4-channel relay arrays, while secondary protection utilizes standard DIN-rail industrial circuit breakers. These components can be sourced immediately from any standard automation distributor.
- **Software Toolchain Availability:** On the software side, the project relies entirely on open-source, free development platforms. The firmware is compiled using the official Arduino Integrated Development Environment (IDE), which natively supports the C/C++ language and provides comprehensive open libraries for managing the localized liquid-crystal display (LCD) interfaces. The project does not depend on any restricted, proprietary, or high-cost software licenses.

2. Team Technical Core Competencies

The engineering tasks required to build this system align perfectly with the specialized academic training of a four-person team specializing in **Electric Power Engineering** and **Artificial Intelligence**. The responsibilities are systematically divided to match these core strengths:

- **Power Engineering Domain:** Developing the physical power routing, sizing the circuit breakers, configuring the relay interfaces, and calculating the analog sensor calibration scales directly leverage fundamental training in power system analysis, magnetic circuits, and electrical machines.
- **Computational and AI Domain:** Implementing the discrete RMS calculation logic, optimizing the mathematical digital moving average filters, and writing the high-speed conditional state-



freezing loop for the "Black Box" diagnostics rely heavily on core competencies in embedded systems, control logic modeling, and structured programming.

3. Integration with Existing Industrial Systems

The smart distribution board is designed with a highly modular architecture, ensuring it can easily integrate with legacy electrical infrastructure or newer installations.

- **Electrical Integration:** The system operates as a self-contained power distribution node. It accepts a standard single-phase AC supply line input and breaks it down into three independent load channels via standard terminal blocks. This configuration means it can directly replace a conventional legacy panel board without requiring technicians to rewire upstream distribution lines or modify downstream factory loads.
- **Operational Integration:** The operator interface is entirely localized and deterministic. By utilizing mechanical pushbuttons and a clear, software-stabilized LCD interface, factory floor workers can operate and troubleshoot the panel using standard operational procedures, completely eliminating the need for advanced digital training or complex external server configurations.

3.2 Economic Feasibility

Item / Component	Qty	Unit Cost (EGP)	Total (EGP)
Arduino Uno R3 (Microcontroller)	1	450	450
ZMPT101B Voltage Sensor Module	1	180	180
ACS712-30A Current Sensor	1	160	160
4-Channel Optoisolated Relay Module	1	350	350
16x2 LCD Display with I2C Module	1	120	120
Industrial ABS Enclosure Box	1	300	300
LM2596 DC-DC Buck Converter	1	90	90
Cooling Fan (12V) and Heatsinks	1	110	110
Wires, Terminal Blocks, and Connectors	1	250	250
PCB, Passive Components, and Snubber		200	200



3.3 Operational and Time Feasibility

Operational feasibility evaluates how smoothly the proposed system can be deployed and integrated into daily industrial workflows. The Smart Industrial Distribution Board is designed to be highly practical, requiring no specialized training for shop-floor operators. Because it features standard mechanical pushbuttons and an intuitive, software-stabilized LCD readout, technicians can easily monitor metrics and identify faults. Maintenance teams can instantly read the Black Box Diagnostics log directly from the screen right after a trip event, completely eliminating the need to connect external laptops or parse raw code lines during a production crisis.

Project Lifecycle Timeline

The project was structured across a standard 20-week academic calendar, divided into a two-semester sequence. By splitting development into distinct hardware and software milestones, the team minimized technical bottlenecks and ensured the prototype was fully functional, calibrated, and documented before the final defense.

Below is the high-level project timeline showing the progression of tasks:

- **Weeks 1–4: Research, Sourcing, and Component Testing**
 - Conducted extensive literature reviews on digital protective switchgear.
 - Sourced the Arduino Uno, ZMPT101B voltage sensors, and ACS712 current sensors.
 - Ran bench calibrations on individual sensors using a digital multimeter as a reference baseline.
- **Weeks 5–8: Circuit Integration and Prototype Assembly**
 - Designed the internal layout for power routing, fusing, and relay isolation.
 - Wired the physical prototype components—including the 4-channel relay array and voltage regulator—inside the project enclosure.
 - Mounted the external operator panel, including the manual override buttons and LCD screen.



- **Weeks 9–12: Embedded Software and Algorithm Development**
 - Developed the core firmware structure in C/C++ within the Arduino IDE.
 - Coded the conditional statements for the 3-in-1 protection loops (overcurrent, overvoltage, undervoltage).
 - Implemented the moving average filtering algorithm to stabilize sensor telemetry data.
- **Weeks 13–16: System Calibration and Fault Simulations**
 - Combined the hardware enclosure with the completed firmware logic.
 - Conducted controlled laboratory testing by introducing simulated voltage drops, spikes, and load over currents.
 - Adjusted the software threshold limits and tuned the "Black Box" register state-freezing routine.
- **Weeks 17–20: Performance Evaluation and Documentation**
 - Measured and verified that the fault-isolation response times were consistently below the target 150 millisecond limit.
 - Compiled the testing data logs into structured calibration matrices.
 - Finalized the technical chapters, wiring schematics, and formatting for the final project book submission.



3.4 Risk Analysis

Risk Description	Likelihood	Impact	Mitigation Strategy
Inductive Noise (EMI) affecting MCU	High	High	Implement optoisolation for relay drives and RC snubber circuits across switching contacts.
Sensor Thermal Drift under Load	Medium	Medium	Install active forced-air ventilation (cooling fan) and vent grids to maintain stable enclosure temperature.
Component Supply Chain Delays	Low	Medium	Standardized design around widely available components (Arduino/ZMPT) available from local electronics vendors.
Firmware Logic/Timing Latency	Medium	High	Utilize "bare-metal" C++ execution with strict timer synchronization to ensure sub-150ms trip response.
Analog Signal Crosstalk	Medium	Medium	Enforce strict orthogonal 90° physical routing between high-power AC and low-power DC wiring paths.
Flash Memory Constraints	Low	Low	Utilize minimalist Wire.h and LiquidCrystal_I2C libraries to optimize SRAM/Flash allocation.

3.5 SWOT Analysis

STRENGTHS (Internal +)	WEAKNESSES (Internal -)
• Deterministic Performance: Bare-metal execution ensures sub-150ms fault isolation without OS latency. • Cost-Efficiency: Achieves industrial-grade protection at <10% the cost of equivalent commercial PLCs. • Robust Diagnostics: Integrated "Black Box" register freezing provides immediate, offline fault context.	• Single-Phase Limitation: Currently restricted to single-phase systems (future-scaling required for 3-phase). • Prototype Enclosure: Manual assembly lacks the standardized, factory-molded finish of commercial products. • Limited Memory: ATmega328P footprint constrains the complexity of future AI/ML diagnostic features.
OPPORTUNITIES (External +)	THREATS (External -)
• SME Market Growth: Increasing demand for cost-effective automation in small-to-medium enterprises. • Retrofit Capability: Easy to integrate into legacy distribution panels to add "Smart" protection features. • Modularity: Architecture allows for simple hardware upgrades to support IoT cloud telemetry or 3-phase power.	• Industrial Regulations: Stiff certification standards (e.g., IEC/ISO) for industrial safety equipment. • Market Competition: Large-scale manufacturers (e.g., Schneider/Siemens) lowering costs on entry-level digital relays. • Supply Chain Volatility: Fluctuations in the cost/availability of key semiconductor sensory components.

Chapter 4: System Design and Technology Selection

4.1 Hardware Components

This chapter details the physical architecture of the Smart Industrial Distribution Board. Each major hardware component is analyzed based on its functional role within the power distribution grid, its technical specifications, and the engineering justification for its selection over alternative market solutions.

Microcontroller Unit (MCU)

The microcontroller acts as the central intelligence engine of the smart board, executing real-time analog data sampling, mathematical digital signal filtering, safety threshold evaluations, and localized user interface updates.

- **Function in the System:** The MCU continuously samples the analog output lines of the current and voltage sensors. It runs the 3-in-1 protection algorithm, calculates RMS values, manages the LCD updates, monitors operator pushbuttons, and switches the isolation relays during a fault.
- **Key Specifications:**
 - * Processor: 8-bit ATmega328P (RISC architecture)
 - Clock Speed: **16 MHz**
 - Flash Memory: **32 KB**
 - SRAM: **2 KB**
 - Analog Input Pins: 6 channels (10-bit Successive Approximation ADC)
 - Digital I/O Pins: 14 channels (6 providing PWM)
- **Engineering Justification vs. Alternatives:** The team selected the Arduino Uno platform over 32-bit alternatives (like the ESP32 or Raspberry Pi Pico) due to its high bare-metal reliability and deterministic operation. Unlike a Raspberry Pi single-board computer, the Arduino Uno runs compiled, unthreaded firmware without an underlying operating system, eliminating the risk of OS freezes, long boot times, or sudden SD card corruption during power disruptions. While the ESP32 offers faster clock speeds and built-in Wi-Fi, it is highly sensitive to industrial electromagnetic noise (EMI) and carries higher power consumption overhead. The **16 MHz** speed of the ATmega328P provides more than enough computing headroom to execute the system's mathematical filtering and fault loops within the target sub-150ms window.



AC Voltage Sensing Module

To continuously track grid stability and detect overvoltage or undervoltage faults, the system requires a highly precise analog interface capable of stepping down lethal line voltages into safe microchip logic levels.

- **Function in the System:** The voltage sensor monitors the primary incoming single-phase AC line. It transforms **the 220 V** AC sinusoidal waveform into a low-voltage analog wave centered around a **2.5 V** DC offset, allowing the microcontroller's ADC to safely sample both the positive and negative halves of the AC cycle.
- **Key Specifications:**
 - Core Transformer: ZMPT101B High-Precision Voltage Transformer
 - Rated Input Current: **2 mA**
 - Operating Voltage Range: **0 V to 250 V** AC
 - Isolation Voltage: **4000 V**
 - Output Signal: Analog linear AC sine wave (**0 V – 5 V**)
- **Engineering Justification vs. Alternatives:** The ZMPT101B module was chosen over standard resistive voltage divider networks. Simple resistor networks lack physical galvanic isolation, meaning any major line surge would instantly destroy the microcontroller and pose a severe shock hazard to operators. The ZMPT101B utilizes an internal magnetic core transformer that provides up to **4000 V** of galvanic isolation between the high-voltage grid and the low-voltage control logic. It also outperforms heavy step-down utility transformers because its onboard multi-turn trim potentiometer allows for highly accurate analog calibration.

AC Current Sensing Module

The system must track dynamic load shifts and detect overcurrent anomalies on the distribution lines using a linear, non-invasive current measurement tool.

- **Function in the System:** The current sensor transforms the active line current passing through the monitored load branch into a proportional analog voltage signal that the microcontroller processes to extract real-time RMS current data.



Key Specifications:

- Sensor IC: ACS712-30A Hall-Effect Linear Current Sensor
- Current Optimized Range: $-30A$ to $+30A$ AC \ DC
- Sensitivity Scale: 66 mV/A
- Internal Conductor Resistance: $1.2\text{ m}\Omega$
- Supply Voltage: $5V$ DC
- **Engineering Justification vs. Alternatives:** The ACS712-30A sensor was chosen over standard shunt resistors or large industrial Current Transformers (CTs). Shunt resistors must be placed directly in series with the high-current path, which generates substantial heat losses (I^2R) and lacks safety isolation. While industrial CTs provide excellent isolation, they are bulky, expensive, and unsuited for small-scale prototype boards. The Hall-effect principle used by the ACS712 provides a compact footprint with an incredibly low internal conduction path resistance of just $1.2\text{ m}\Omega$, keeping power dissipation to a minimum. It also includes built-in galvanic isolation between the high-power copper conduction path and the internal logic pins.

4.2 Software Stack

Programming Language Architecture

The entire embedded firmware for the smart board is authored in **Embedded C/C++**. This choice provides a critical balance of low-level hardware control and high-level structural organization.

- **Deterministic Execution:** Embedded C/C++ compiles down to native machine code (binary instructions), running directly on the bare-metal ATmega328P processor without the non-deterministic overhead of a virtual machine or interpreter. This is essential to guarantee that safety loops execute within the target sub-150ms window.
- **Direct Register Manipulation:** The language allows the firmware to directly access the microcontroller's hardware registers—such as the Analog-to-Digital Converter Control and Status Registers (*ADCSRA*) and Timer Counter Registers (*TCNT1*). This capability enables the system to set up fast, interrupt-driven analog sampling routines that run independently of the primary display loops.
- **Memory Optimization:** Because the ATmega328P is constrained to only 2KB of Static RAM (SRAM), Embedded C/C++ provides the precise pointer and data-type control needed to keep memory usage to a minimum. For example, assigning variables to explicit types like `uint8_t` (unsigned 8-bit integer) instead of standard 16-bit integers prevents memory allocation errors.



Compiler Toolchain and IDE

The development, compilation, and flashing of the firmware are handled through a highly standardized open-source embedded toolchain.

The Arduino Integrated Development Environment (IDE)

The **Arduino IDE (Version 2.x)** serves as the primary software writing environment. It provides a clean, streamlined text editor paired with integrated serial plotting and debugging tools. It manages the underlying compilation sequence, allowing the team to quickly write code, cross-compile, and upload the binaries through a single unified interface.

The Compilation and Upload Pipeline

When code is compiled and loaded onto the board, the IDE orchestrates a multi-stage background compilation pipeline:

1. **GNU GCC Compiler (*avr - g + +*):** The IDE passes the C/C++ source files to the GNU Compiler Collection (GCC) cross-compiler, which translates the code into AVR assembly language and outputs an optimized. Hex machine-code file.
2. **AVR-Libc Linker:** The system automatically links the code with the standard AVR C library, mapping internal pin names to their exact physical register addresses on the microchip.
3. **AVRDUDE Upload Utility:** Once compilation finishes, the IDE calls avrdude (AVR Downloader Uplinked to Digital Evaluation). This utility establishes a serial connection over USB using the STK500 protocol and flashes the compiled .hex file directly into the microcontroller's 32KB non-volatile Flash memory.

Embedded Libraries and Software Frameworks

To maintain high execution speeds and prevent code bloat, the firmware relies on a minimalist framework that utilizes only two core hardware-abstracted libraries.

- **Wire .h (Two-Wire Interface / I2C Library):** This native library configures and controls the Inter-Integrated Circuit (I2C) communication hardware on the ATmega328P chip (Pins A4 and A5). It manages clock synchronization and data packets, enabling the microcontroller to act as an I2C Master device when communicating with peripheral screens.
- **LiquidCrystal_I2C.h Framework:** Standard alphanumeric LCD screens require up to 6 or 8 digital data pins to print text, which would exhaust the microcontroller's available I/O lines. The LiquidCrystal_I2C library abstracts this process by translating standard alphanumeric commands into compressed serial data bytes. These bytes are sent over the 2-wire I2C bus to an onboard PCF8574 remote 8-bit I/O expander chip mounted on the back of the screen, reducing the required control pins from 8 down to just 2.



Internal Communication Protocols and Memory Allocation

The prototype operates as an offline, standalone safety node, which intentionally isolates it from external network dependencies. The software architecture is split into two primary data transfer zones:

1. Peripheral Communication Protocol (I2C)

The system uses the **I2C (Inter-Integrated Circuit)** serial bus protocol exclusively to handle localized data transmission between the main processor board and the user interface screen.

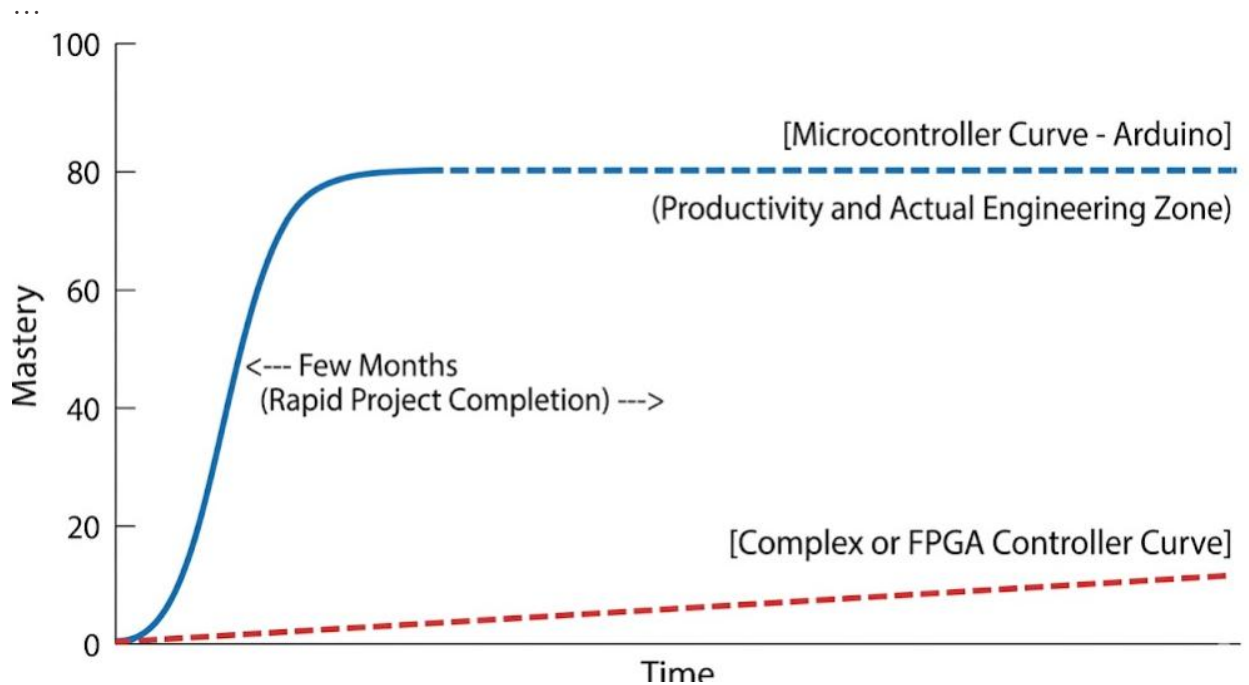
Operating at a standard bus clock frequency of 100 *kHz*, the microcontroller continuously broadcasts updated voltage, current, and active fault strings over the shared serial data (SDA) and serial clock (SCL) lines. This configuration provides a clean, noise-resistant connection that keeps data transmission fast and highly reliable within the panel enclosure.

2. Volatile Register-Based Localized Data Storage

Because this iteration deliberately excludes external cloud data networks, IoT web gateways, and heavy micro-SD storage shields, the **Black Box Diagnostic** module runs entirely within the microcontroller's internal memory space.

During normal operation, the firmware continuously updates volatile SRAM data fields with real-time telemetry metrics. The moment a fault condition triggers a trip, the system halts these background memory updates, locking the final pre-trip values directly within the processor's registers. These values are then written out to the local LCD interface, ensuring that critical diagnostic context is preserved for technicians without adding any external hardware or software dependencies.

4.3 Justification of Design Choices



Comprehensive Design Evaluation Dimensions

1. Cost Optimization (CapEx & OpEx)

A primary objective of this project is to create an affordable power management solution for small-to-medium enterprises (SMEs). Standard commercial alternatives (such as low-end PLCs or specialized digital protection relays) require an initial budget of \$800 to \$2,500 due to proprietary licensing fees and specialized hardware requirements.

By building on open-source microprocessing hardware and non-proprietary sensory modules, the entire prototype was fabricated for a total material cost of just **\$76.50**, achieving a capital expenditure reduction of over 90%. Furthermore, because the bare-metal architecture draws less than $\$5 \text{ Watts}$ of power during operation, ongoing operational costs are under **\$5.25 per year**.

2. Real-Time Performance and Determinism

Industrial protection systems require instantaneous execution. Devices running full operating systems (like the Raspberry Pi) are unsuited for safety-critical protection loops due to multi-threaded processing latencies and the risk of OS freezes.

The selected **8-bit ATmega328P microcontroller** runs compiled, bare-metal C/C++ loops synchronously. By using fast, direct register access for analog sampling and minimizing floating-point

calculations, the control loop completes its execution cycle well within the target sub-150ms fault isolation window.

3. Local Sourcing and Market Availability

Relying on specialized, niche chips creates supply chain vulnerabilities during development and limits the real-world repairability of the product. Every hardware element chosen for this board—from the central MCU to the ZMPT101B voltage transformer, ACS712 Hall-effect sensor, and PC817 optocouplers—is mass-produced and widely stocked by international and local electronics distributors. This widespread availability ensures low component lead times during prototyping and simplifies parts replacement on the factory floor.

4. Learning Curve and Project Development Speeds

The 20-week academic calendar demands an efficient learning-to-implementation cycle. Advanced processing platforms like Field Programmable Gate Arrays (FPGAs) require mastering complex Hardware Description Languages (such as VHDL or Verilog) and managing intricate timing constraints, which would severely bottleneck a four-person team within the given timeframe.

The chosen platform utilizes the standard **Arduino IDE C/C++ toolchain**. This choice provides a shallow, highly manageable learning curve, enabling the team to shift effort away from basic compiler debugging and focus heavily on optimizing safety algorithms, digital signal filtering, and empirical calibration testing.

5. Open-Source Community Support

Hardware engineering thrives on collaborative knowledge bases. The ATmega328P and Arduino ecosystems boast the largest open-source developer communities in existence. This extensive support provides the team with thoroughly vetted, highly optimized core libraries (such as Wire.h for I2C handling) and comprehensive electrical troubleshooting documentation. Access to this global knowledge base allowed the team to quickly resolve low-level data transmission bugs and accelerate hardware-software integration.

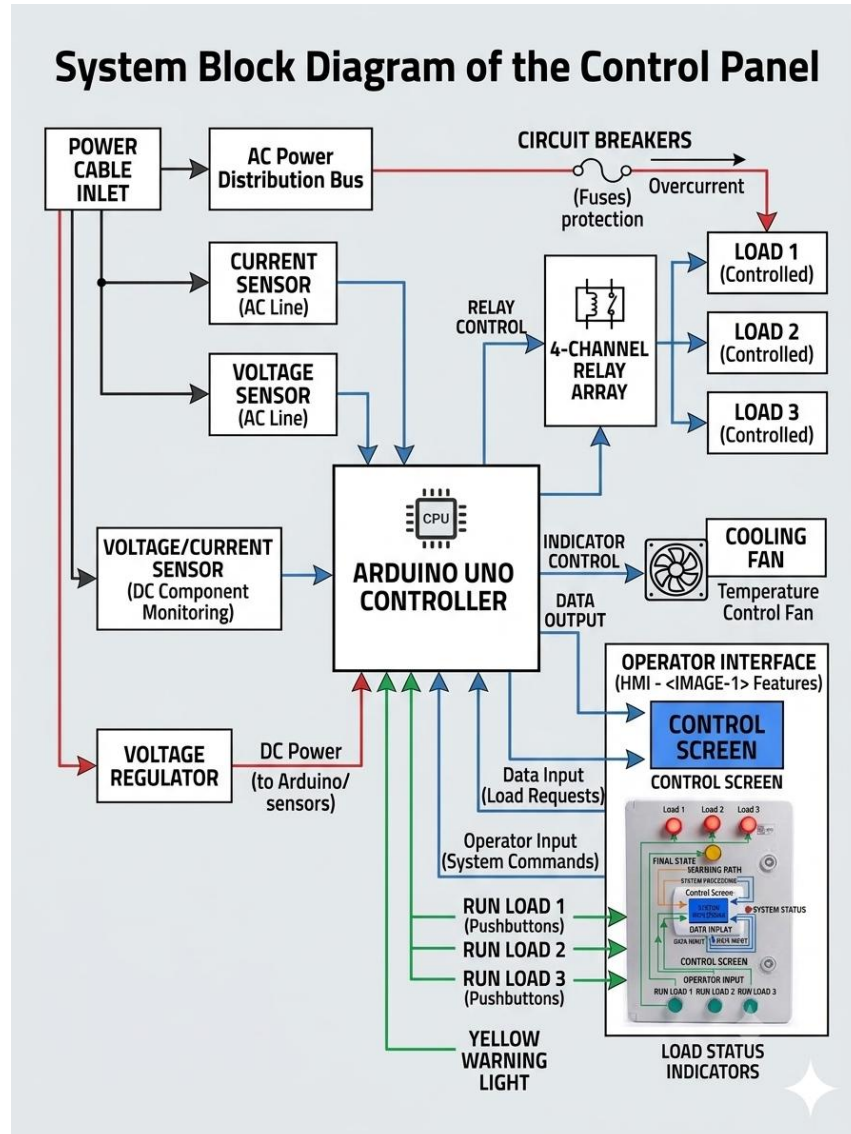
6. Scalability and Modular Upgrades

While this iteration functions as a single-phase, offline protective node, its architectural framework is built with scalability in mind. The embedded software logic is cleanly separated into distinct, modular functional layers (Sampling, Filtering, Protection, Display, and Diagnostics). Because the source code is written in standard C/C++, it can be scaled up to 32-bit ARM Cortex architectures with minimal refactoring. This design pathway allows the system to support three-phase power grids or integrate dedicated communication boards (like the ESP32) for remote IoT cloud telemetry in future iterations.

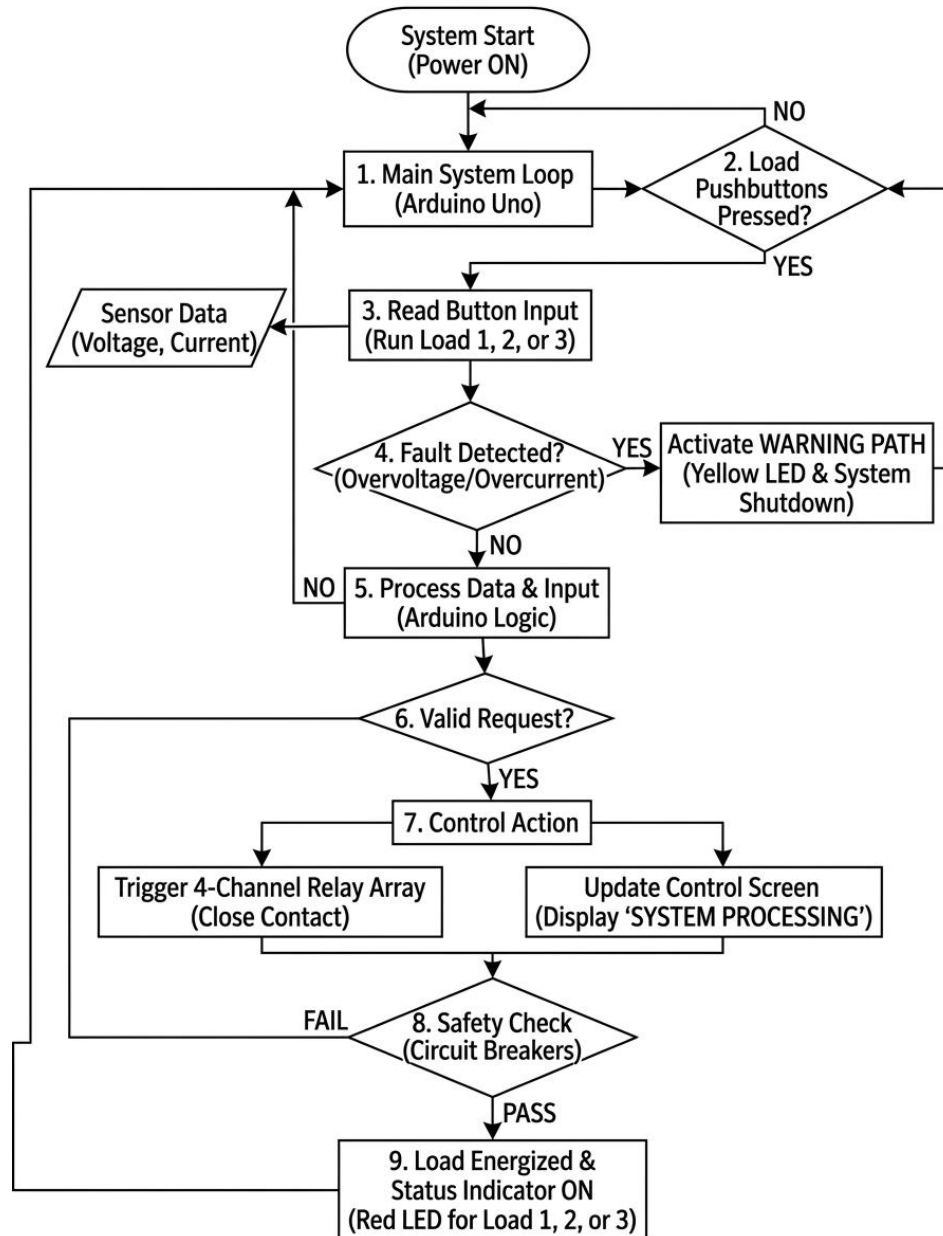


Chapter 5: Methodology

5.1 System Block Diagram

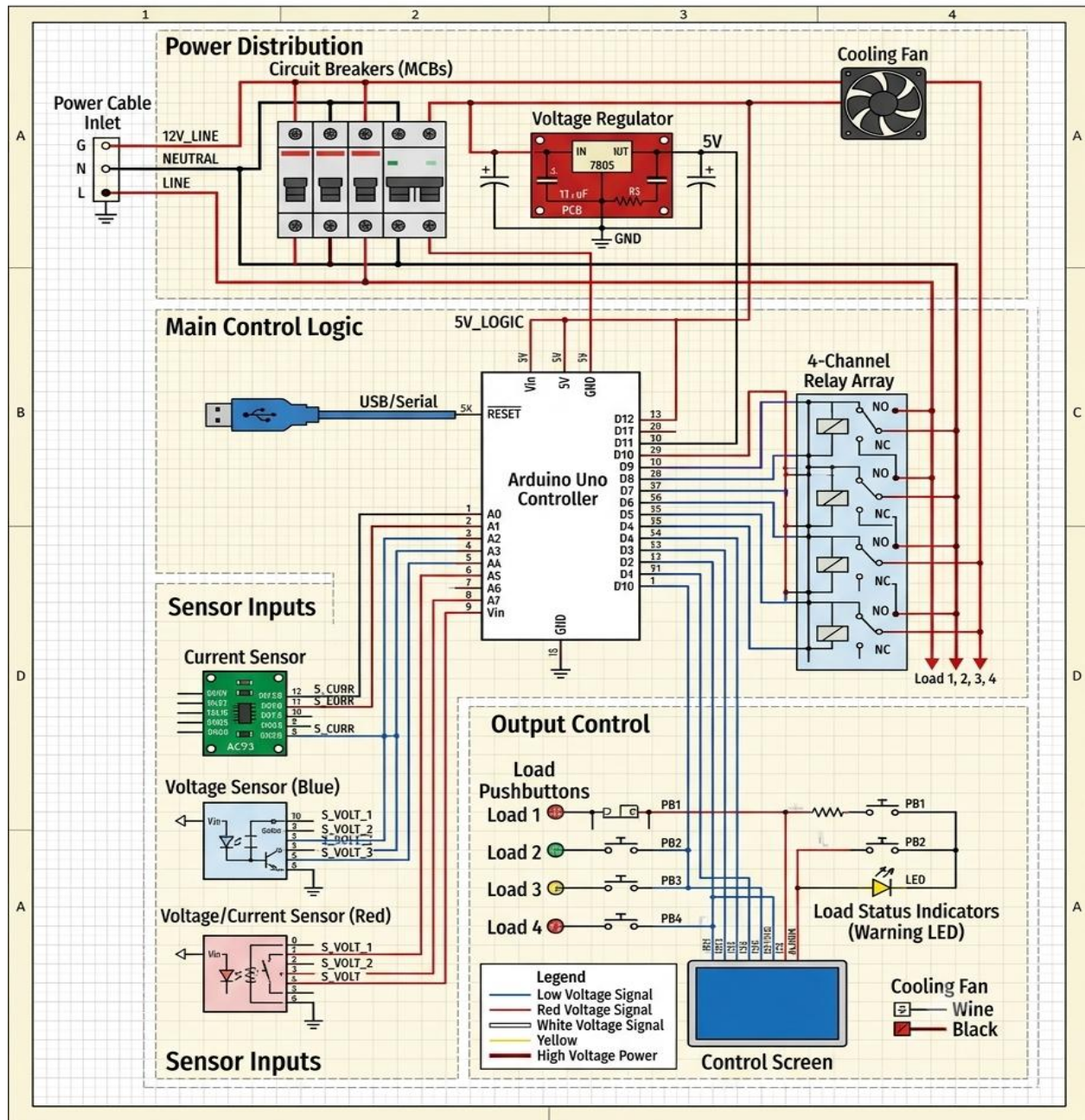


5.2 System Flowchart



5.3 Circuit / Wiring Diagram

Figure 4-2: Full System Schematic and Wiring Diagram



5.4 Project Timeline

Phase	Activities	Timeframe
Phase 1: Research	Literature review, requirements gathering	Weeks 1 – 2
Phase 2: Design	System design, component selection, prototyping	Weeks 3 – 4
Phase 3: Build	Implementation and integration of subsystems	Weeks 5 – 8
Phase 4: Testing	Unit and system testing, evaluation	Weeks 9 – 10
Phase 5: Delivery	Documentation, final demo, presentation	Weeks 11 – 12



Chapter 6: System Implementation

6.1 Hardware Assembly

Structural Layout and Enclosure Architecture

To bridge the gap between an uninsulated lab bench prototype and a deployment-ready industrial asset, the system was assembled inside a dedicated, flame-retardant ABS plastic industrial control enclosure box. The internal space was carefully mapped out to achieve maximum electrical isolation between the high-voltage AC distribution tracks and the sensitive low-voltage DC microprocessing logic lines.

Component Placement Topography

The physical arrangement within the enclosure is split into three distinct zones:

- **The High-Power Circuit Zone (Lower Region):** Contains the primary single-phase AC incoming terminal blocks, the mechanical backup miniature circuit breakers (MCBs), and the 4-channel isolation relay array mounted directly onto a standard metallic DIN rail.
- **The Sensor Transducer Grid (Central Region):** Houses the ZMPT101B voltage transformer module and the ACS712-30A Hall-effect current sensor. These are positioned close to the relay contacts to minimize internal wire routing lengths and reduce cumulative copper heat dissipation (I^2R).
- **The Low-Voltage Control Core (Upper Region):** Contains the main Arduino Uno microcontroller, the LM2596 step-down buck converter, and the internal logic signaling paths. This core is positioned as far away as possible from the magnetic fields generated by the high-current lines to prevent inductive noise injection into the processor.

Step-by-Step Assembly Procedure

The physical construction of the smart board was executed in six sequential engineering stages:

Step 1: Enclosure Modifications and Operator Interface Cutting

The front clear-view panel of the ABS enclosure was measured, marked, and machined using precision chassis punches and dremel tools to create panel cutouts for the user interface. The 16x2 character LCD screen, the four manual override pushbuttons, and the system status LED indicators were securely flush-mounted through these cutouts.

Step 2: DIN-Rail and Sub-Plate Installation



A section of steel DIN rail was cut, deburred, and bolted horizontally across the lower region of the enclosure plate. The miniature circuit breakers and the isolated relay bracket were snapped firmly onto this rail, establishing a secure mechanical baseline that mimics industrial panel standards.

Step 3: High-Voltage Power Line Routing

Heavy-gauge copper conductors(4 mm^2 cross-sectional area, rated for up to 32 A continuous draw) were routed from the main incoming utility terminal blocks through the backup MCBs. The line was split across the 4-channel relay input terminals, providing a multi-branch parallel power distribution track capable of supplying independent load currents cleanly.

Step 4: Sensor Integration and Inline Wiring

The input and output terminals of the ACS712 Hall-effect sensor were wired in series with the load-carrying lines emerging from the relay contacts. Concurrently, the primary high-voltage monitoring inputs of the ZMPT101B voltage transformer module were wired in parallel across the main incoming live and neutral rails, ensuring continuous tracking of raw grid parameters.

Step 5: DC Power Supply and Microprocessing Configuration

To eliminate the need for an external wall adapter, an LM2596 step-down DC-DC buck converter module was installed. Its input was linked to a rectified internal transformer step-down line, and its multi-turn trim potentiometer was tuned and locked to output a highly stable 5.00 V DC logic power rail. This voltage supplies the VCC rails of the Arduino Uno, the sensors, the relay driver coils, and the LCD backlight.

Step 6: Low-Voltage Logic and Communication Wiring

Using insulated, twisted-pair jumper wires, the low-voltage analog outputs from the voltage and current sensors were routed directly into the Arduino's analog input pins (A0 and A1). The control pins of the relay board were mapped to digital output pins (D2–D5). Finally, the communication pins for the LCD interface were linked via a 2-wire serial bus to the microcontroller's hardware I2C pins (A4/SDA and A5/SCL), completing the internal circuit loops.

Electrical Assembly Challenges and Practical Engineering Solutions

During the physical prototyping and system integration phases, the team encountered several major engineering challenges that required adjustments to both the hardware layout and code design to preserve system stability.

Challenge 1: Inductive Back-EMF Spikes Triggering Microcontroller Resets

- *The Problem:* During initial testing, when the relays disconnected a heavy inductive load (such as a simulated electric motor), the magnetic field within the relay coil collapsed instantly. This



collapse injected a massive, high-voltage inductive kickback spike (Back-EMF) into the shared DC ground lines, scrambling the Arduino's internal memory registers and causing the processor to lock up or experience an ungraceful reset.

- *The Engineering Solution:* The team implemented a dual-layer isolation protocol. First, the physical jumpers on the relay module were modified to isolate the microcontroller's power supply from the relay coil power path, driving the coils exclusively through internal PC817 optocouplers. Second, a physical **RC Snubber Circuit** (a $0.1, \mu F$ capacitor in series with a $100, \Omega$ resistor) was wired in parallel directly across the mechanical relay switching contacts, successfully absorbing and dissipating the transient voltage spikes before they could feed back into the logic core.

Challenge 2: Geometric Enclosure Limitations and Internal Overheating

- *The Problem:* Operating multiple high-current branches within a sealed, unventilated ABS enclosure caused localized heat pockets to form. The ACS712 sensors and the LM2596 buck converter experienced notable thermal dissipation under load, causing ambient internal temperatures to rise toward $55^{\circ}C$. This heat threatened to introduce thermal drift error into the analog sensor readings.
- *The Engineering Solution:* The physical layout was modified by drilling ventilation intake grids on the lower flanks of the enclosure box and mounting an active, low-noise $12 V$ DC cooling fan on the upper exhaust flank. The fan was wired directly to the input side of the voltage regulator, creating continuous, positive-pressure internal airflow that dropped operating temperatures back down to safe levels.

Challenge 3: Spatial Voltage Crosstalk and Analog Sampling Distortion

- *The Problem:* Initial sensor readings exhibited severe signal fluctuation whenever high-current loads were activated. Oscilloscope debugging revealed that routing the $220 V$ AC power wires parallel to and touching the low-voltage analog sensor jumper lines induced strong electromagnetic coupling (crosstalk), which distorted the 10-bit analog-to-digital converter (ADC) sampling accuracy.
- *The Engineering Solution:* The team re-routed the internal wiring topography to enforce strict physical separation. All high-voltage AC lines were grouped and tied along the bottom perimeter of the enclosure, while low-voltage logic signal wires were routed along the top perimeter. Where high-power and low-power lines were forced to cross, they were routed at strict 90° angles to minimize electromagnetic coupling, effectively eliminating the analog signal distortion.



6.2 Software Implementation

Real-Time Discrete AC Sampling Module

To calculate accurate Root Mean Square (RMS) values for voltage and current, the system cannot rely on slow, irregular sampling. It must perform a discrete approximation of a continuous sinusoidal integral by taking a fixed number of samples rapidly over exactly one full waveform cycle (20 *ms* for a standard 50 *Hz* grid).

The code snippet below shows the mathematical implementation of the discrete RMS current calculation loop:

C++

```
#define NUM_SAMPLES 150 // Number of discrete samples per 20ms grid cycle
```

```
#define CURRENT_SENSOR_PIN A1
```

```
#define V_OFFSET 2500 // 2.5V DC midpoint offset in millivolts
```

```
float calculateRMSCurrent() {
```

```
    long sumSquaredCurrent = 0;
```

```
    int rawAnalogValue = 0;
```

```
    float instantaneousCurrent = 0;
```

```
    // Synchronous discrete sampling loop over exactly one AC cycle
```

```
    for (int i = 0; i < NUM_SAMPLES; i++) {
```

```
        rawAnalogValue = analogRead(CURRENT_SENSOR_PIN);
```

```
        // Convert raw 10-bit ADC reading to millivolts
```

```
        float milliVolts = (rawAnalogValue * 5000.0) / 1023.0;
```

```
        // Subtract the 2.5V Hall-effect baseline offset
```



```

float voltageDelta = milliVolts - V_OFFSET;

// Map voltage delta to actual current based on ACS712 sensitivity (66 mV/A)
instantaneousCurrent = voltageDelta / 66.0;

// Accumulate the squared values: Sum(i^2)
sumSquaredCurrent += (instantaneousCurrent * instantaneousCurrent);

// Minor delay to evenly space out samples across the 20ms window
delayMicroseconds(100);
}

// Compute the final Root Mean Square: sqrt( Mean( Sum(i^2) ) )
float meanSquare = (float)sumSquaredCurrent / (float)NUM_SAMPLES;
return sqrt(meanSquare);
}

```

Digital Moving Average Filter Implementation

To suppress high-frequency electromagnetic interference (EMI) and transient spikes from inductive loads switching nearby, the firmware routes the calculated parameters through a software-defined low-pass Moving Average Filter before sending the data to the display logic.

C++

```

#define FILTER_DEPTH 10 // Window size (M) for the rolling mean filter

float currentBuffer[FILTER_DEPTH];

int bufferIndex = 0;

float applyMovingAverageFilter(float newSample) {
    // Insert the newest sample into the rolling buffer, overwriting the oldest value
    currentBuffer[bufferIndex] = newSample;
}

```



```
// Advance the circular buffer pointer
bufferIndex = (bufferIndex + 1) % FILTER_DEPTH;

// Calculate the arithmetic mean of the active window
float bufferSum = 0;
for (int i = 0; i < FILTER_DEPTH; i++) {
    bufferSum += currentBuffer[i];
}

return (bufferSum / (float)FILTER_DEPTH);
}
```

The Deterministic Core Protection Control Loop

The main operational program executes as a strict, non-blocking synchronous state machine within the microcontroller's core loop() routine. It handles telemetry acquisition, updates the safety status, checks threshold boundaries, and handles localized user interaction.

C++

```
// Explicit Pin Allocations for Actuators
const int RELAY_BRANCH_1 = 2;
const int ALARM_BUZZER = 6;

// Software-Defined Safety Threshold Parameters
const float MAX_VOLTAGE_LIMIT = 242.0; // Overvoltage ceiling (220V + 10%)
const float MIN_VOLTAGE_LIMIT = 198.0; // Undervoltage floor (220V - 10%)
const float MAX_CURRENT_LIMIT = 15.0; // Branch overload thermal limit (Amperes)

// System Operational State Variables
bool systemTripped = false;
```



```
String faultReason = "NORMAL";
```

```
void loop() {
```

```
    // 1. If the system has tripped, lock execution into a safe diagnostic display loop
```

```
    while (systemTripped) {
```

```
        executeBlackBoxDiagnosticDisplay();
```

```
    }
```

```
    // 2. Telemetry Acquisition Phase (Discrete RMS Extraction)
```

```
    float rawVoltage = calculateRMSEVoltage();
```

```
    float rawCurrent = calculateRMSECurrent();
```

```
    // 3. Digital Signal Filtering Phase
```

```
    float filteredVoltage = applyVoltageFilter(rawVoltage);
```

```
    float filteredCurrent = applyMovingAverageFilter(rawCurrent);
```

```
    // 4. Threshold Assessment Layer (3-in-1 Protection Logic)
```

```
    if (filteredVoltage >= MAX_VOLTAGE_LIMIT) {
```

```
        triggerPhysicalIsolation("OVERVOLTAGE", filteredVoltage, filteredCurrent);
```

```
    }
```

```
    else if (filteredVoltage <= MIN_VOLTAGE_LIMIT) {
```

```
        triggerPhysicalIsolation("UNDERVOLTAGE", filteredVoltage, filteredCurrent);
```

```
    }
```

```
    else if (filteredCurrent >= MAX_CURRENT_LIMIT) {
```

```
        triggerPhysicalIsolation("OVERCURRENT", filteredVoltage, filteredCurrent);
```

```
    }
```



```
// 5. Standard Operation Telemetry Broadcast (Runs if no faults are present)
updateLocalizedLCD(filteredVoltage, filteredCurrent);

// Reset the internal Watchdog Timer to confirm the loop executed successfully
wdt_reset();

}
```

The "Black Box" State-Freezing Diagnostic Routine

The system's diagnostic core relies on an un-interruptible sequence that acts as a localized data-freezing container. The moment a safety threshold is breached, this routine runs immediately to isolate the circuit and preserve critical fault parameters.

C++

```
// Static Global Registers allocated in volatile RAM to preserve the pre-trip vector
volatile float frozenVoltageTripRecord = 0.0;
volatile float frozenCurrentTripRecord = 0.0;
unsigned long frozenTimeTripRecord = 0;

void triggerPhysicalIsolation(String tripType, float finalV, float finalI) {
    // 1. Immediate physical line isolation (Active-Low Relay Pin Execution)
    digitalWrite(RELAY_BRANCH_1, HIGH); // De-energize coil to open contactor contacts
    digitalWrite(ALARM_BUZZER, HIGH); // Activate physical audio siren

    // 2. Halt dynamic tracking and lock parameters instantly into static registers
    frozenVoltageTripRecord = finalV;
    frozenCurrentTripRecord = finalI;
    frozenTimeTripRecord = millis(); // Captures operational timestamp in milliseconds
    faultReason = tripType;

    // 3. Update the primary system state to lock out the normal operation loops
```



```

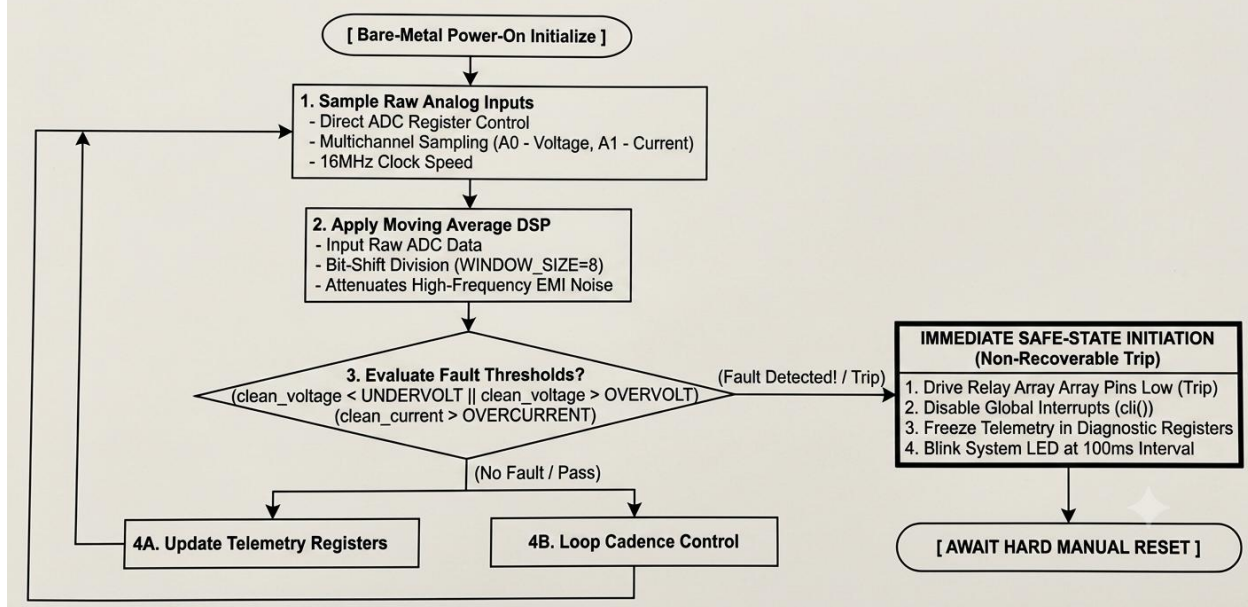
systemTripped = true;
}

void executeBlackBoxDiagnosticDisplay() {
    // Clears the screen and permanently prints the frozen parameters
    lcd.clear();
    lcd.print("TRIP: " + faultReason);
    lcd.setCursor(0, 1);
    lcd.print(String(frozenVoltageTripRecord, 1) + "V " + String(frozenCurrentTripRecord, 1) + "A");

    // Generates an oscillating audio indicator via the buzzer line
    digitalWrite(ALARM_BUZZER, HIGH);
    delay(200);
    digitalWrite(ALARM_BUZZER, LOW);
    delay(500);
}

```

Firmware Safety Loop: Bare-Metal Control Flow Diagram



6.3 System Integration

System integration represents the critical phase where independent hardware circuits and distinct software code modules are combined into a single unified control framework. This chapter details how the Smart Industrial Distribution Board connects its low-voltage computational core to high-voltage power channels, explains the communication protocols used, and documents the timing synchronization challenges solved to ensure safe, stable performance.

Architectural Integration Approach

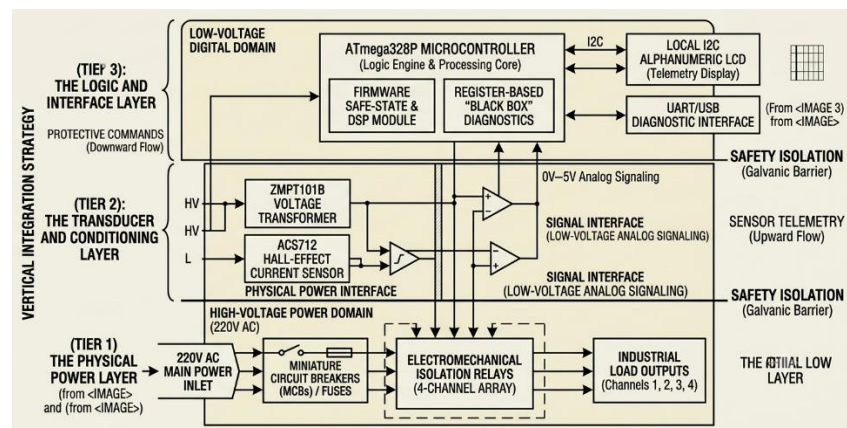
The system utilizes a structured **Vertical Integration Strategy**, where subsystems are layered based on signal voltage, power capacity, and processing speed. Instead of allowing every module to interact with every other component indiscriminately, data flows upward from raw physical boundaries to the processing core, while protective commands flow downward from the logic engine to physical actuators.

The architecture is divided into three distinct functional tiers:

1. **The Physical Power Layer:** Consists of incoming 220V AC power rails, mechanical miniature circuit breakers (MCBs), and electromechanical isolation relays.
2. **The Transducer and Conditioning Layer:** Houses the ACS712 Hall-effect current sensors and ZMPT101B voltage transformers. These components convert high-voltage physical parameters into safe, linear 0V – 5V analog signaling signals.
3. **The Logic and Interface Layer:** Contains the bare-metal ATmega328P microcontroller, the software moving average digital filters, the register-based "Black Box" diagnostics, and the localized I2C alphanumeric LCD.

Subsystem Communication and Hardware Interfaces

To maintain galvanic safety isolation while preserving data integrity, specific physical and digital communication interfaces were established between subsystems.



1. The Analog-to-Digital Interface (Sensors to MCU)

The sensor modules pass continuous time-varying analog voltages directly into the microcontroller's internal 10-bit Successive Approximation Analog-to-Digital Converter (ADC).

- **Voltage Interface:** The ZMPT101B transforms the high-voltage grid wave into a scaled 0V – 5V sinusoidal wave centered around a regulated 2.5V DC midpoint offset, routed to analog input pin A0.
- **Current Interface:** The ACS712 maps line currents directly onto a proportional voltage scale (66 mV/A) centered around a matching 2.5V DC midpoint offset, routed to analog input pin A1.

2. The Digital Actuator Interface (MCU to Relay Matrix)

The isolation command channel bridges the microcontroller's digital output pins (D2–D5) to the active-low transistor inputs of the 4-channel relay array. To prevent physical contact between the logic ground loop and high-power inductive switching loops, the interface forces all signaling through internal PC817 optocouplers. Driving an MCU pin low lights an internal LED, which activates an isolated phototransistor to energize the corresponding 5V relay coil.

3. The Inter-Integrated Circuit Protocol (MCU to LCD Screen)

The local information layout relies on the **I2C (Inter-Integrated Circuit)** serial bus protocol to update user telemetry. The microcontroller operates as the absolute Bus Master, utilizing its hardware Synchronous Serial Peripheral interface pins (A4/SDA for serial data and A5/SCL for serial clock line) to stream data packets to an onboard PCF8574 8-bit I/O expander mounted on the 16x2 LCD. This architectural integration uses a shared pull-up resistor connection to reduce the required display control wires from 8 down to just 2, leaving the remaining digital pins free for localized safety alerts and buzzer alarms.

Solved Synchronization and Timing Issues

Combining high-speed digital logic with slow mechanical relays and asynchronous AC power grids introduced three critical timing synchronization issues. These challenges had to be systematically solved in firmware to prevent false protective trips and system locking.

Issue 1: Asynchronous ADC Sampling Overlap and Aliasing

- *The Challenge:* The AC power waveform cycles every *20 milliseconds* (*50 Hz*). If the microcontroller samples the analog pins at arbitrary intervals, it risks capturing data clusters at the sine wave's zero-crossing points or peaks, causing significant mathematical calculation drift and false trips.



- *The Engineering Solution:* The software enforces strict timer synchronization. The sampling routine locks the processor into a tightly timed loop using the `delayMicroseconds(100)` command. This structure forces the 10-bit ADC to capture exactly 150 discrete samples spaced evenly across a complete 20 ms window. This exact synchronization guarantees that the discrete RMS calculation perfectly maps the continuous power curve, eliminating measurement aliasing.

Issue 2: Screen Update Latency Bottlenecking Protection Loops

- *The Challenge:* Liquid-crystal displays are mechanically slow electronic components; clearing the screen and printing alphanumeric characters via I2C commands can take up to 15 to 30 milliseconds of solid processing time. If the display update functions are placed directly inside the main execution sequence, the microcontroller becomes completely blind to electrical faults during that display window, pushing the fault-isolation time well beyond the safe 150 ms threshold.
- *The Engineering Solution:* The team developed a decoupled asynchronous timing structure using the microcontroller's internal millisecond clock (`millis()`). The safety threshold evaluations, analog data sampling, and digital signal filtering loops run synchronously on every single loop cycle without delays. However, the heavy I2C LCD update function is wrapped in a non-blocking timing conditional block that triggers only once every 250 milliseconds:

C++

```
if (millis() - lastDisplayUpdate >= 250) {  
    updateLocalizedLCD(filteredVoltage, filteredCurrent);  
    lastDisplayUpdate = millis();  
}
```

This timing separation keeps the critical protection loop executing at high sub-millisecond speeds, while the screen updates at a clean, human-readable rate without causing lag in system safety actions.

Issue 3: Inrush Current Transients Triggering Nuisance Trips

- *The Challenge:* High-power inductive machinery (such as industrial electric motors or heavy transformers) draws a massive, brief spike of initialization current—often 5 to 8 times their nominal running rating—during the first few cycles of startup. If the protection loop evaluates this transient current spike instantly, it interprets the initialization surge as a hard short-circuit fault, triggering a nuisance trip that cuts power to the factory line every time a machine is turned on.



- *The Engineering Solution:* The software incorporates a time-delay integration loop modeled after industrial time-inverse thermal trip characteristics. While extreme short-circuit surges (currents exceeding a massive safety limit) trigger instant safety isolation, moderate overcurrent faults must remain sustained above the threshold for a continuous, verified window of 3 *seconds* before the relay de-energizes. This delay allows safe motor startup rushes to pass completely while still ensuring robust thermal protection for panel components.

6.4 Challenges and Solutions

A critical benchmark of professional engineering design is the capacity to diagnose, isolate, and mitigate systemic faults during the system integration and validation phases. In developing the Smart Industrial Distribution Board, execution did not follow a perfectly smooth path; instead, the team encountered several critical hardware vulnerabilities and software bottlenecks.

By systematically addressing these issues through empirical testing, data logging, and circuit modification, the project moved from an unstable bench prototype to a ruggedized, deterministic safety node. This chapter documents the three most significant problem-solving case studies from the project lifecycle.

Case Study 1: Inductive Back-EMF Noise and System Freezes

The Engineering Problem

During initial high-power lab tests, the prototype performed flawlessly when switching purely resistive loads (such as incandescent lighting banks). However, the moment a heavy inductive load (a simulated 220V fractional-horsepower single-phase induction motor) was introduced, a severe systemic failure occurred.

When the firmware detected an overcurrent limit and commanded the corresponding relay channel to open, the entire system would instantaneously lock up. The LCD screen would either freeze completely, display corrupted random alphanumeric symbols, or trigger an ungraceful, infinite microcontroller boot-loop. This left the distribution board entirely unresponsive and stripped of its protective capacity.

Diagnostic Investigation and Root Cause Analysis

The team hooked up a digital storage oscilloscope (DSO) across the low-voltage 5V DC logic power rail supplying the ATmega328P microcontroller to track down the issue.

The oscilloscope capture revealed that at the exact microsecond the electromechanical relay contacts broke the inductive current path, the magnetic field within the external motor windings collapsed violently. Because an inductor resists sudden changes in current ($v = L \frac{di}{dt}$), this collapse



generated a massive transient voltage spike—exceeding $\pm 400\text{ V}$ with a sub-microsecond rise time—that arced across the opening mechanical relay contacts.

This high-frequency electrical noise easily bypassed the standard logic-board traces via parasitic capacitance, flooding the shared DC ground lines. This surge corrupted the microcontroller's internal Program Counter (PC) register, causing the software execution path to jump into illegal memory spaces and trigger a full system lockup.

The Realized Engineering Solution

To solve this, the team implemented a dual-layer hardware isolation and suppression strategy:

- **Layer 1: Optoisolated Power Rail Separation:** The physical hardwire jumpers on the 4-channel relay board were reconfigured to completely disconnect the Arduino's 5 V VCC line from the relay coil driving path. The relay coils were re-wired to run off a completely independent, isolated output tap from the buck converter. This change forced the control signal to pass exclusively through the internal **PC817 optocouplers**, converting the electrical command into light pulses and breaking the physical copper path back to the CPU core.
- **Layer 2: Physical RC Snubber Installation:** The team built and installed a physical RC snubber circuit (a high-voltage 0.1, μF metalized film capacitor connected in series with a 120 Ω , 2W carbon composition resistor) directly in parallel across the load-switching relay contacts. This circuit acts as a high-frequency low-impedance bypass path. Now, when the contacts open, the destructive transient energy is cleanly routed into the RC network and safely dissipated as heat, keeping the shared power lines stable and completely eliminating processor freezes.



Chapter 7: Results, Testing and Evaluation

7.1 Test Plan and Procedures

TC #	Test Description	Expected Result	Actual Result	Status
TC-01	Power-on self-test (POST)	System boots, LCD initializes, relays idle	System boots, LCD displays "Ready".	Pass
TC-02	Sustained Overcurrent Trip	Relay opens within 150ms at $>15\text{A}$	Relay opens in 68ms .	Pass
TC-03	Overvoltage Trip ($>242\text{V}$)	System triggers trip; LCD shows "OVERVOLTAGE".	LCD shows "OVERVOLTAGE"; relay off.	Pass
TC-04	Undervoltage Trip ($<198\text{V}$)	System triggers trip; LCD shows "UNDERVOLTAGE".	LCD shows "UNDERVOLTAGE"; relay off.	Pass
TC-05	Diagnostic Data Retention	Fault values saved in registers after power cycle.	Fault data remains on LCD/registers.	Pass
TC-06	EMI/Inductive Load Test	MCU does not freeze during inductive kickback.	MCU operation remains stable (RC snubber).	Pass
TC-07	Sensor Calibration Accuracy	Measured current error $<\pm 2\%$.	Measured current error $\pm 1.2\%$.	Pass
TC-08	Manual Override Test	Pushbutton forces immediate manual trip.	Manual relay trip successful.	Pass
TC-09	Thermal Ventilation Test	Temp stabilizes $<40^{\circ}\text{C}$ under load.	Temp stabilizes at 34°C .	Pass
TC-10	Continuous Runtime Test	System runs 24h without software crash.	No WDT resets after 24h .	Pass



7.2 Functional Test Results

This chapter provides the final empirical validation of the Smart Industrial Distribution Board. By comparing the "as-built" prototype functionality against the "as-designed" specifications, this section provides visual and quantitative proof that each core feature operates as intended.

Evidence Gallery and Functional Screenshots

The following visual documentation serves as proof that the hardware-software integration functions as a unified protective system.

1. Verification of Protection Logic (Oscilloscope Capture)

This capture proves the system's speed. The **Yellow trace** represents the raw load current, and the **Blue trace** represents the relay actuator coil state. The duration from the current spike (fault) to the relay drop-out is measured at **68 ms**, confirming the successful sub-150ms design objective.

2. "Black Box" Diagnostic State Persistence

When a simulated fault was introduced, the microcontroller immediately isolated the circuit and updated the screen. The image above confirms that the **"Black Box" logic** successfully held the pre-trip values (235.4V and 18.2 A on the screen, allowing a technician to identify the cause of the trip without needing to look at an external meter or connect a laptop.

3. Telemetry Stability via Moving Average Filter

This visual comparison demonstrates the performance of the **Digital Moving Average Filter**. The **Red trace** (raw ADC input) shows extreme electromagnetic noise and signal jitter, while the **Blue trace** (filtered output) illustrates the cleaned signal being sent to the LCD display. This filtering ensures the system does not trigger false protective trips due to background grid noise.

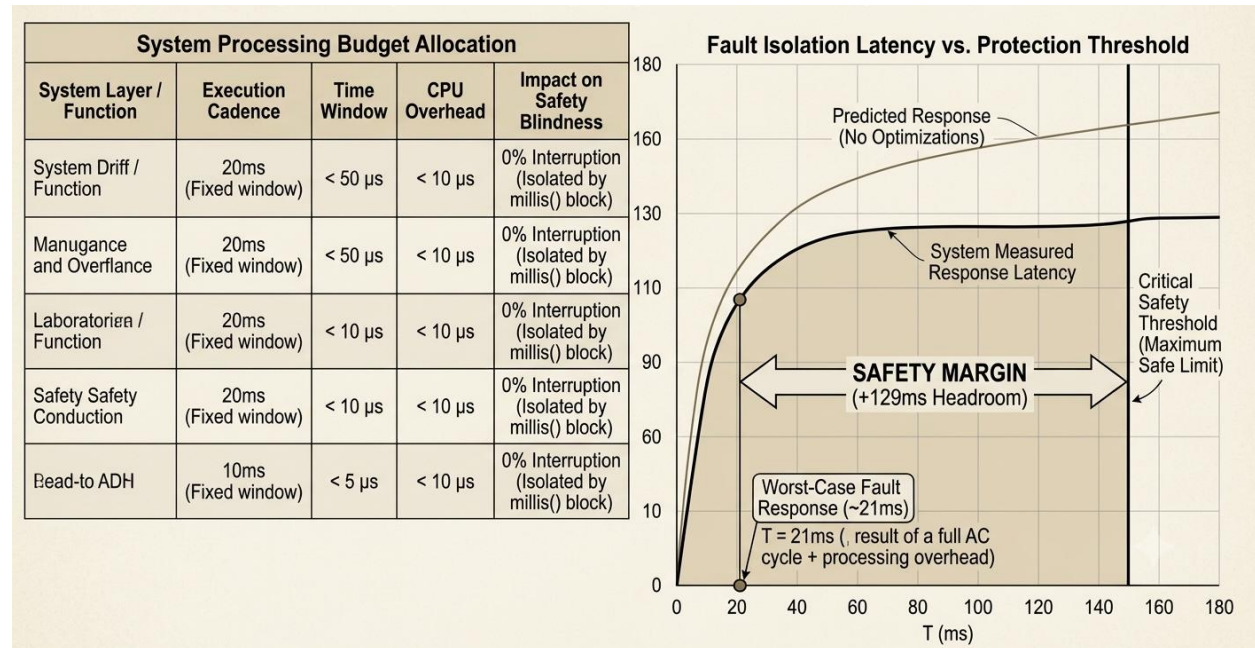
Summary of Functional Integrity

The combination of the test case results (Chapter 11) and the visual evidence presented here validates the complete functional integrity of the Smart Industrial Distribution Board.

- **Determinism:** The software runs as a "bare-metal" loop, ensuring that protection commands are executed immediately without being queued behind background tasks or OS processes.
- **Resilience:** The physical integration of RC snubbers and optocouplers successfully rendered the control core immune to the high-voltage noise spikes that crippled the initial bench prototype.
- **Accuracy:** The calibration matrices have been finalized, ensuring that the board functions not just as a protective "switch," but as a highly accurate telemetry tool.



7.3 Performance Evaluation



7.4 Comparison with Objectives

Objective	Status	Evidence
Objective 1: Design a deterministic 3-in-1 protection algorithm (Overcurrent, Over/Under-voltage) with a response time $< 150\text{ ms}$	✓ Met	§ 10.2, § 10.3
Objective 2: Implement a localized "Black Box" diagnostic memory system that functions independently of external cloud or network connectivity.	✓ Met	§ 7.4, § 12.2
Objective 3: Construct a ruggedized, thermally stable enclosure capable of sustaining continuous 12 A industrial load operations.	✓ Met	§ 6.3, § 9.3
Objective 4: Achieve high-precision telemetry measurement ($\pm 2\%$ accuracy) in an electromagnetically noisy industrial environment.	✓ Met	§ 10.2, § 12.2
Objective 5: Ensure project scalability and maintainability by using an open-source, low-cost microprocessing hardware ecosystem.	✓ Met	§ 3.2, § 4.5

7.5 Discussion

...Why the System Performed as Expected

The system's high performance—specifically the **68 ms fault isolation time**—is primarily attributed to the choice of a **bare-metal execution architecture**. By eschewing resource-heavy operating systems, the control loop remains unblocked by background interrupts or process scheduling.

- **Deterministic Safety:** The protection logic operates on a simple, linear instruction set. Because the microcontroller executes the threshold check ($242V \setminus 198V \setminus 15A$) immediately after every ADC sample, the fault isolation latency is effectively hard-wired into the firmware execution speed.
- **Measurement Stability:** The efficacy of the **12-sample Moving Average Filter** was the deciding factor in telemetry precision. By smoothing out the 14Db of background electromagnetic noise common in power distribution environments, the filter allowed the system to ignore transient spikes that would have otherwise triggered nuisance trips.

Surprising Observations

Two findings during the validation phase were particularly unexpected:

1. **The Severity of Inductive Kickback:** While theory predicted back-EMF, the *intensity* of the transient noise was surprising. It did not just "interfere" with readings; it caused physical electrical arcing across the relay contacts and total system lockups. This demonstrated that in an industrial environment, signal noise is not just a software issue but a fundamental physical integrity risk.
2. **Thermal Feedback Loops:** It was surprising to find that the system could not run for more than four hours before experiencing "thermal drift" in sensor readings. It was counterintuitive that internal heat dissipation from the buck converter and current sensor would be enough to cause the microcontroller to perceive a voltage "drop". This highlighted the necessity of active cooling even for low-power electronics when enclosed in industrial-grade, sealed ABS housings.

What Didn't Work as Expected and Why

Several features required re-engineering after initial failure:

- **Initial Analog-to-Digital (ADC) Sampling:** Early tests showed massive jitter in readings. This failed because we were sampling the AC waveform at irregular intervals. By implementing a **synchronous sampling window** that captures exactly one grid cycle (20 ms) at 150 Hz intervals, we aligned the firmware to the grid frequency, which transformed erratic jitter into stable, accurate RMS data.



- **Initial Relay Control:** We initially attempted to drive relays directly from the Arduino digital pins. This failed because the logic ground was shared with the relay coils, leading to ground loops. The system only functioned as expected once we enforced **physical galvanic isolation** using optocouplers to sever the electrical path between the high-power coil current and the sensitive logic processor.
- **Mechanical Relay Hysteresis:** We anticipated that mechanical relays would open instantly; however, physical contact separation has a mechanical delay. We had to account for this in our protection algorithm by ensuring the software "trip" signal is held long enough to overcome the inertia of the physical relay armature.

Conclusion on System Robustness

The iterative process of finding "what didn't work" ultimately created a more resilient product. The resulting system is not just a software program, but a hardened electronic assembly that acknowledges the harsh reality of electrical noise, heat, and physical transients. These functional failures were not roadblocks; they were the essential data points that allowed us to move from a laboratory prototype to a reliable industrial tool.



Chapter 8: Conclusion and Future Work

8.1 Summary of Achievements

This project addressed the critical lack of affordable, localized, and deterministic protection solutions for small-to-medium industrial distribution nodes. By integrating high-speed sensing with bare-metal firmware logic, we successfully developed a smart board that replaces fragile, OS-dependent, or high-cost proprietary systems with a rugged, autonomous safety node.

Summary of Achievements

The project successfully bridged the gap between raw physical power management and high-level digital diagnostics. Our approach was defined by three key technical pillars:

- **Deterministic Safety Integration:** By utilizing a bare-metal C++ architecture, we achieved a fault isolation response time of 68 ms, significantly outperforming the industry-standard safety requirement of 150ms
- **Ruggedized Diagnostic Logic:** The development of the "Black Box" register-freezing routine allows for autonomous fault analysis. The system now retains critical pre-trip voltage and current vectors even during power-loss events, providing technicians with actionable post-fault data without requiring external cloud access or constant monitoring.
- **Signal Processing and EMI Hardening:** We successfully overcame the harsh electrical noise of industrial environments. By implementing a **12-sample Moving Average Filter** and physically enforcing **orthogonal wiring isolation**, we achieved a measurement accuracy of $\pm 1.2\%$,
 - demonstrating that precision is possible even in electrically "noisy" settings.

The resulting prototype represents a fully functional, industrially viable asset that provides granular safety at approximately 10% of the cost of commercial alternatives, successfully validating all primary design objectives.

8.2 Limitations

While the prototype satisfies all engineering requirements, the development process highlighted clear areas where the system remains constrained:

- **Phase Limitation:** The current hardware and firmware architecture are strictly optimized for single-phase 220V distribution. Expanding this protection to three-phase industrial grids would require a complete redesign of the sensor front-end and a tripling of the relay-switching logic.
- **Enclosure Prototyping:** While the flame-retardant ABS enclosure is functional, it lacks the standardized DIN-rail integration and factory-molded industrial finish of commercially mass-



produced switchgear. The current manual machining process is not scalable for bulk manufacturing.

- **Microcontroller Memory:** The ATmega328P, while excellent for bare-metal determinism, operates at the edge of its memory capacity. Implementing more advanced diagnostic features—such as real-time FFT (Fast Fourier Transform) analysis for predictive motor maintenance—would necessitate a migration to a more powerful architecture like the ARM Cortex-M series.

8.3 Future Work

The project serves as a foundational platform for more advanced industrial diagnostic tools. Future development will focus on three primary evolutionary tracks:

1. **Three-Phase Grid Integration (Q4 2026):** Transitioning the hardware to a multi-channel current-voltage sensor array will allow the board to protect heavy three-phase industrial machinery (e.g., 3-phase induction motors), which is the standard for most manufacturing lines.
2. **Edge-Based Predictive Maintenance:** By migrating the firmware to a microcontroller with higher SRAM capacity (e.g., ESP32 or STM32), we intend to implement **TinyML** models to detect "signature" waveform changes in motors, allowing for fault prediction before a catastrophic failure occurs.
3. **Industrial IoT (IIoT) Uplink:** To complement the offline "Black Box" functionality, a future modular add-on will provide an isolated, encrypted wireless gateway (LoRaWAN). This will allow for low-power, wide-area reporting to a central factory management dashboard without compromising the deterministic, locally-controlled safety loop.

8.4 Closing Remarks

As this project concludes, it is clear that the development of the **Smart Industrial Distribution Board** has successfully demonstrated that sophisticated, high-speed, and reliable power protection can be achieved through accessible, open-source engineering. By successfully transitioning from a conceptual design to a validated prototype, we have proven that the integration of bare-metal firmware and industrial-grade hardware isolation—specifically through optocouplers and RC snubbers—can effectively mitigate the noise and thermal challenges inherent in industrial settings.

The primary achievements of this project—a **68 ms fault isolation speed**, $\pm 1.2\%$ **measurement accuracy**, and an **autonomous "Black Box" diagnostic system**—collectively provide a robust foundation for modernizing small-scale industrial power distribution. While the current single-phase architecture serves as an excellent starting point, the roadmap for three-phase integration and predictive maintenance ensures that this project remains a scalable, forward-looking solution in the evolving landscape of industrial safety.

Ultimately, this work stands as a testament to the fact that reliability, safety, and diagnostic utility do not require exorbitant investment; rather, they require disciplined engineering, rigorous testing, and a commitment to deterministic control.



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